

# Data Preprocessing and Feature Engineering Report

## Stroke Prediction Dataset

### 1. Dataset Overview

The Stroke Prediction dataset was loaded using Pandas. The dataset contains patient demographic, medical, and lifestyle information. The target variable is **stroke**, which indicates whether a patient has experienced a stroke.

### 2. Data Quality Assessment

Missing values were checked for all columns. Duplicate records were identified and removed where applicable. Data types were verified. Categorical values were inspected for inconsistencies. Outliers were detected using the IQR method and visualized using box plots.

### 3. Data Preprocessing

Missing BMI values were handled using imputation. Duplicate records were removed. Data types were verified and categorical values were standardized. Outliers were handled using an appropriate preprocessing method to improve data quality.

### 4. Feature Engineering

Three new features were created: **BMI\_Category** (groups BMI into health categories), **Age\_Group** (categorizes patients by age), and **Hypertension\_Risk** (combines hypertension and heart disease information). These features provide more meaningful information for machine learning models.

### 5. Feature Handling

The dataset was divided into input features (X) and target variable (y). Categorical variables were encoded using Label Encoding and One-Hot Encoding. Numerical features were scaled using both StandardScaler and MinMaxScaler. StandardScaler standardizes values around zero, while MinMaxScaler scales values to the range 0–1.

### 6. Final Dataset

The final dataset is cleaned, feature-engineered, encoded, and scaled, making it suitable for training machine learning models such as Logistic Regression, Decision Tree, Random Forest, and other classification algorithms.

### 7. Conclusion

The preprocessing and feature engineering steps improved the overall quality of the dataset by handling missing values, duplicates, inconsistent data, and outliers. The engineered features and transformed variables provide a reliable dataset for building and evaluating stroke prediction models.